ERD

Entities

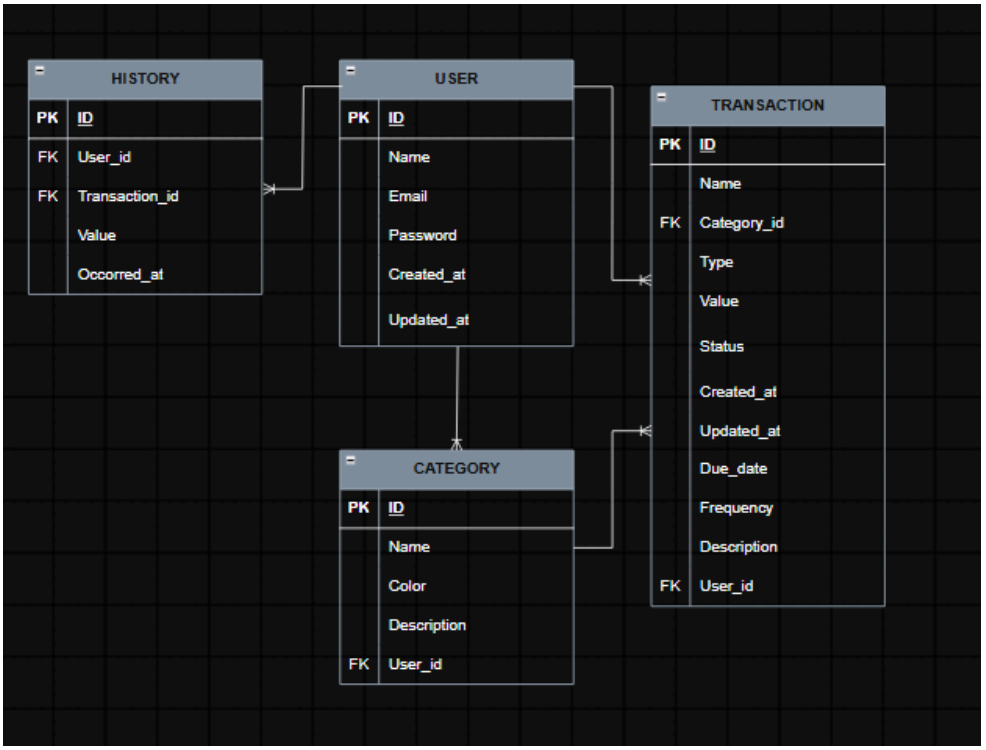
1. User
    a. ID
    b. name
    c. email
    d. password
    e. created_at
    f. updated_at
2. Transaction
    a. ID
    b. name
    c. user_id
    d. category_id
    e. type
        i. in
        ii. out
    f. value
    g. status
        i. pending
        ii. received
        iii. paid
    h. created_at
    i. updated_at
    j. due_date
    k. frequency
        i. once
        ii. monthly
        iii. weekly
    l. description
3. Category
    a. ID
    b. user_ID
    c. name
    d. color
    e. description
4. History
    a. ID
    b. user_id
    c. transaction_id
    d. value

e. occorred_at

Relationship:

1. User **1** → **N** Transaction: have
2. User **1** → **N** Category: have
3. Category **1** → **N** Transaction: has transactions
4. User **1** → **N** History: creates history

Diagram:



Attribute Constraints:

1. Type: **in** (income) / **out** (expense)
2. Status: pending, paid, received
3. Frequency: once, weekly, monthly

SQL CODE

```
CREATE TYPE transactions_type AS ENUM (
'IN',
'OUT'
);

CREATE TYPE transactions_status AS ENUM (
'RECEIVED',
'PENDING',
'PAID'
);

CREATE TYPE transactions_frequency AS ENUM (
'ONCE',
'MONTHLY',
'WEEKLY'
);

CREATE TABLE history (
id SERIAL PRIMARY KEY,
users_id INT NOT NULL,
transactions_id INT NOT NULL,
value NUMERIC(10,2) NOT NULL,
occurred_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE categories (
id SERIAL PRIMARY KEY,
users_id INT NOT NULL,
name VARCHAR(30) NOT NULL,
color VARCHAR(7) NOT NULL,
description VARCHAR(50)
);

CREATE TABLE transactions (
id SERIAL PRIMARY KEY,
name VARCHAR(30) NOT NULL,
categories_id INT NOT NULL,
users_id INT NOT NULL,
type transactions_type NOT NULL,
value DECIMAL(10,2) NOT NULL,
status transactions_status NOT NULL,
frequency transactions_frequency NOT NULL DEFAULT 'ONCE',
```

```
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP,
due_date DATE,
description VARCHAR(50)
);

CREATE TABLE users (
id SERIAL PRIMARY KEY,
name VARCHAR(50) NOT NULL,
email VARCHAR(255) NOT NULL UNIQUE,
password_hash VARCHAR(255) NOT NULL,
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP
);

ALTER TABLE transactions
ADD CONSTRAINT fk_transactions_users
FOREIGN KEY (users_id)
REFERENCES users(id),
ADD CONSTRAINT fk_transactions_categories
FOREIGN KEY (categories_id)
REFERENCES categories(id),
ADD CONSTRAINT chk_transactions_type_status
CHECK (
   (type = 'IN' AND status = 'RECEIVED')
   OR
   (type = 'OUT' AND status IN ('PENDING', 'PAID'))
),
ADD CONSTRAINT uq_transactions_id_users
UNIQUE (id, users_id);

ALTER TABLE history
ADD CONSTRAINT fk_history_transaction_user
FOREIGN KEY (transactions_id, users_id)
REFERENCES transactions (id, users_id),
ADD CONSTRAINT uq_history_transaction
UNIQUE (transactions_id);

ALTER TABLE categories
ADD CONSTRAINT fk_categories_users
FOREIGN KEY (users_id)
REFERENCES users(id);
```